Zhang Hanbo

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Education

National University of Singapore — Singapore
Visiting Ph.D. Student (Virtually), Supervised by David Hsu — *Apr. 2020 – Current*

Xi'an Jiaotong University — Xi'an, China
Ph.D. Candidate of Control Science and Engineering, Supervised by Xuguang Lan — *Aug. 2016 – Current*

Xi'an Jiaotong University — Xi'an, China
Bachelor of Information Engineering — *Aug. 2012 – Jun. 2016*

Professional Experience

Visiting Ph.D. Student — Apr. 2020 – Current
National University of Singapore — *School of Computing*

- *Interactive grasping following unrestricted language instructions*: My research aims to enable the robot to follow unrestricted human's instruction for goal-directed object retrieval by grasping. I proposed INVIGORATE algorithm, the abbreviation for INteractive VIsual GrOunding and gRAsping in clutTEr. By integrating learning and planning, INVIGORATE formulates the problem as a POMDP on top of the deep visual models and achieves robust performance for natural-language-guided goal-directed grasp planning in clutter scenes.

Ph.D. Student — Sep. 2016 – Current
Xi'an Jiaotong University — *Institute of Artificial Intelligence and Robotics, IAIR*

- *Robotic grasp detection*: My research aims to help the robot perform grasping robustly while intelligently. I proposed a series of robotic grasp detection algorithms, to help the robot: 1) grasp unknown objects in single-object scenes robustly; 2) grasp specified objects in clutter; 3) perform task-oriented grasping; 4) detect robust 6D grasp configurations in an end-to-end manner in clutter taking as input point clouds.
- *Visual manipulation relationship*: My research aims to generate the safe grasping plan in clutter. The grasping order is represented by the so-called manipulation relationship. I proposed Visual Manipulation Relationship Network to detect manipulation relationships visually.
- *Reinforcement learning and robot control*: My research aims to efficiently learn the robot control policy. I proposed a RL algorithm called Hindsight Trust Region Policy Optimization (HTRPO), which theoretically inherits the impressive converging property of the well-known TRPO and adapts to the sparse-reward settings. It is promising in the future to enable the end-to-end grasping control learning by taking RGB images as inputs.

Undergraduate Research Assistant — Sep. 2015 – Jun. 2016
Xi'an Jiaotong University — *Institute of Artificial Intelligence and Robotics, IAIR*

- *Robotic grasp detection*: Implemented a learning-based and model-free robotic grasp detection algorithm following *Lenz et al. IJRR, 2013*.
- *Object Detection*: Implemented an object detector based on *Deformable Part Model*.
- *Robot grasping system*: Designed a vision-based robotic grasping system to grasp objects without explicit models on ROS and a Baxter robot.

Publications

- **Zhang, H.**, Lu, Y., Yu, C., Hsu, D., Lan, X., Zheng, N. INVIGORATE: Interactive Visual Grounding and Grasping in Clutter. Robotics: Science and Systems (RSS), 2021.
- **Zhang, H.**, Bai, S., Lan, X., Hsu, D., Zheng, N. Hindsight Trust Region Policy Optimization. arXiv preprint arXiv:1907.12439. International Joint Conference of Artificial Intelligence (IJCAI), 2021.

- Zhao, B., **Zhang, H.**, Lan, X., Wang, H., Tian, Z., Zheng, N. REGNet: REgion-based Grasp Network for Single-shot Grasp Detection in Point Clouds. IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2021
- **Zhang, H.**, Lan, X., Zhou, X., Tian, Z., Zhang, Y., Zheng, N. Visual Manipulation Relationship Recognition in Object-Stacking Scenes. Pattern Recognition Letters, vol: 140, pp: 34-42, 2020
- Yang, C., Lan X., **Zhang, H.**, Zheng, N. Autonomous Tool Construction with Gated Graph Neural Network. IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2020
- **Zhang, H.**, Zhou, X., Lan, X., Li, J., Tian, Z., Zheng, N., A Real-Time Robotic Grasping Approach With Oriented Anchor Box. In IEEE Transactions on Systems, Man, and Cybernetics: Systems. doi: 10.1109/TSMC.2019.2917034, 2019
- **Zhang, H.**, Lan, X., Bai, S., Zhou, X., Tian, Z., Zheng, N. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2019
- **Zhang, H.**, Lan, X., Wan, L., Yang, C., Zhou, X., Zheng, N. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2019
- Yang. C., Lan, X., **Zhang, H.**, Zheng, N. Task-oriented Grasping in Object Stacking Scenes with CRF-based Semantic Model. 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2019
- **Zhang, H.**, Lan, X., Zhou X., Tian, Z., Zhang, Y., Zheng, N. Robotic Grasping in Multi-object Stacking Scenes Based on Visual Reasoning. SCIENTIA SINICA Technologica, vol: 48(12), pp: 1341-1356. 2018
- **Zhang, H.**, Lan, X., Zhou, X., Tian, Z., Zhang, Y., Zheng, N. Visual Manipulation Relationship Network for Autonomous Robotics. IEEE-RAS 18th International Conference on Humanoid Robots (HUMANOIDS) (pp. 118-125). IEEE, 2018
- Zhou, X., Lan, X., **Zhang, H.**, Tian, Z., Zhang, Y., Zheng, N. Fully Convolutional Grasp Detection Network with Oriented Anchor Box. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 7223-7230). IEEE, 2018
- Yang, C., Lan, X., **Zhang, H.**, Zhou, X., Zheng, N. Visual Manipulation Relationship Detection with Fully Connected CRFs for Autonomous Robotic Grasp. IEEE International Conference on Robotics and Biomimetics (ROBIO) (pp. 393-400). IEEE, 2018
- **Zhang, H.**, Lan, X., Zhou, X., Wang, J., Zheng, N. Vision-based Robotic Grasp Success Determination with Convolutional Neural Network. IEEE 7th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER) (pp. 31-36). IEEE, 2017, **Best Paper Finalist.**

Awards

3nd of IROS 2020 OCRTOC, Simulation Contest — *IEEE, 2020*
Special Academic Scholarship — *Xi'an Jiaotong University, 2020*
2nd of Tri-Co Robot Challenging — *NSFC, 2019*
IEEE CYBER Best Paper Finalist — *IEEE, The USA, 2017*
National Encouragement Scholarship — *MOE of P.R.C., 2015*

Professional Activities

- **Reviewer**
  Conferences: IROS 2021, ICRA 2021, IROS 2020, AIM 2020
  Journals: IEEE Trans. SMC, IEEE Access